

EduVision_DV – Higher Education Performance Dashboard

Project Documentation

1. Project Overview

EduVision_DV is a Higher Education Performance Dashboard developed to analyze and visualize global university performance, research activity, student characteristics, academic indicators, and country-level education trends.

The project combines data from the **QS World University Rankings 2025** and **Times Higher Education (THE) World University Rankings 2016–2026** to create an integrated analytical dataset.

The final solution is presented through four interconnected Tableau dashboards:

1. **University Overview**
2. **Research Analytics**
3. **Student Analytics**
4. **Country Comparison**

The project demonstrates an end-to-end data analytics and business intelligence workflow, from data preparation and KPI development to interactive visualization and benchmarking.

2. Project Objectives

The primary objectives of **EduVision_DV** are:

- To analyze global university performance.
- To compare universities using ranking and performance indicators.
- To analyze research publications, citations, productivity, and impact.
- To examine student population and international student representation.
- To analyze faculty-to-student relationships.
- To study student diversity and enrollment patterns.
- To compare university performance across countries and regions.

- To identify higher-education trends over time.
- To develop meaningful KPIs for university benchmarking.
- To provide an interactive and user-friendly Tableau dashboard.

3. Project Scope

The project covers the following major areas:

- University ranking analysis
- Academic performance analysis
- Research analytics
- Student analytics
- Internationalization analysis
- Faculty and student analysis
- Country benchmarking
- Regional education trends
- Performance Index analysis
- Interactive dashboard development

The project is intended to provide a consolidated view of higher education performance rather than focusing on a single university or country.

4. Data Sources

4.1 QS World University Rankings 2025

The QS World University Rankings 2025 dataset provides university-level ranking and location information.

The dataset contributes to:

- University identification
- Ranking comparison
- Institutional analysis

- Geographic analysis
- Academic performance analysis

4.2 Times Higher Education World University Rankings 2016–2026

The THE World University Rankings dataset contains university information covering the period from **2016 to 2026**.

It provides information related to:

- University rankings
- Research
- Teaching
- Academic reputation
- Faculty
- Student population
- International students
- International outlook
- Research quality
- Research environment
- Overall performance

The multi-year structure also enables trend analysis.

5. Data Preparation Methodology

The project follows a structured data analytics workflow:

Data Collection → Data Cleaning → Data Integration → KPI Development → Performance Analysis → Dashboard Development → Testing & Validation

Step 1 – Data Collection

The QS and THE university-ranking datasets are collected and prepared for analysis.

Step 2 – Data Inspection

The datasets are examined to understand:

- Number of records
- Available fields
- Data types
- Missing values
- Duplicate records
- Numerical and categorical variables

Step 3 – Data Cleaning

The datasets are cleaned to improve consistency and usability.

Major cleaning activities include:

- Removing duplicate records.
- Standardizing university names.
- Standardizing country names.
- Removing unnecessary spaces.
- Inspecting missing values.
- Preparing numerical fields.
- Converting percentage fields into usable numerical values.
- Converting student population ranges into analytical values.

Step 4 – Dataset Integration

The QS and THE datasets are integrated using standardized university and country information.

This creates a consolidated dataset suitable for further analysis.

Step 5 – KPI Development

Important performance indicators are developed to support:

- University analysis

- Research analysis
- Student analysis
- Country comparison

Step 6 – Performance Analysis

Selected indicators are normalized to make them suitable for comparative analysis.

A project-specific **Performance Index** is then developed.

Step 7 – Dashboard Development

The final analytical dataset is used to create four Tableau dashboards.

Step 8 – Testing and Validation

The dashboard calculations, filters, trends, comparisons, maps, and navigation are reviewed before final delivery.

6. Key Performance Indicators

The project contains several important analytical indicators.

KPI	Purpose
Global Rank Score	Supports comparative university ranking analysis
Research Productivity Index	Measures research quality relative to the available faculty/student-related indicator
Faculty-to-Student Ratio	Measures the relationship between faculty and students
International Student Percentage	Measures international student representation
Academic Reputation	Supports academic excellence analysis
Overall Score	Supports university performance comparison
Publications	Measures research output
Citations	Measures research influence

Research Impact	Supports research-impact analysis
Student Population	Measures enrollment scale
Student Diversity	Supports diversity analysis
Performance Index	Provides an overall project-specific benchmarking measure

7. Performance Index

The project introduces a **Performance Index** to support comparative university analysis.

The index combines three normalized components:

Component	Weight
Global Rank Score	40%
Research Productivity Index	30%
International Student Percentage	30%

Interpretation

A higher Performance Index indicates stronger performance according to the selected indicators and weights used in this project.

Important: The Performance Index is a **project-specific analytical measure**. It is not an official QS or THE ranking score.

8. Tableau Dashboard Suite

The final project consists of four interconnected dashboards.

9. Dashboard 1 – University Overview

The **University Overview** dashboard provides a high-level view of global university performance.

9.1 KPIs

- Total Universities
- Total Countries

- Total Regions
- Average Academic Reputation
- Average Overall Score

9.2 Visualizations

Top University Rankings

Displays universities with strong ranking/performance values.

Global University Distribution

Shows the geographical distribution of universities across regions.

Employment Outcomes by Universities

Provides a comparison of universities based on employment outcomes.

University Performance Trends

Shows changes in university performance over time.

Institutional Comparison

Allows users to compare institutions using selected performance indicators.

9.3 Purpose

The dashboard provides a quick overview of:

- University performance
- Academic reputation
- Geographic distribution
- Employment outcomes
- Institutional comparisons

10. Dashboard 2 – Research Analytics

The **Research Analytics** dashboard focuses on research performance and research impact.

10.1 KPIs

- Total Publications
- Total Citations
- Average Citations
- Average Publications
- Average Research Impact

10.2 Visualizations

Publications Analysis

Compares universities based on publication output.

Citation Performance

Analyzes citation performance across universities or regions.

Top Research Institutions

Highlights institutions with strong research performance.

Research Productivity Trends

Shows changes in research productivity over time.

Research Impact Comparison

Compares institutions using research-related indicators.

10.3 Purpose

This dashboard helps users understand:

- Research output
- Research influence
- Citation performance
- Research productivity
- Research-performing institutions

11. Dashboard 3 – Student Analytics

The **Student Analytics** dashboard focuses on student characteristics and institutional student-related indicators.

11.1 KPIs

- Total Students
- International Students
- Faculty-to-Student Ratio
- Average Student Diversity
- Total Faculty

11.2 Visualizations

International Student Analysis

Compares international student representation.

Faculty-to-Student Ratio Analysis

Examines the relationship between faculty and students.

Enrollment Comparisons

Compares universities based on student population.

Student Diversity Trends

Shows changes in student diversity over time.

Student Distribution Analysis

Shows the distribution of students across locations.

11.3 Purpose

The dashboard helps analyze:

- Student population
- International students

- Diversity
- Faculty/student relationships
- Enrollment patterns

12. Dashboard 4 – Country Comparison

The **Country Comparison** dashboard provides country-level and regional benchmarking.

12.1 KPIs

- Total Countries
- Average Overall Score
- Average Global Rank Score
- Average Performance Index
- Total Universities

12.2 Visualizations

Country Ranking Comparison

Compares countries based on university-ranking representation.

Education Performance Benchmarking

Compares countries using selected performance indicators.

Regional Education Trends

Shows education-performance trends across regions.

Top Performing Countries

Highlights countries with strong average performance.

12.3 Purpose

The dashboard supports:

- Country benchmarking
- Regional comparison

- Education performance analysis
- Identification of high-performing countries
- Trend analysis

13. Dashboard Interactivity

The Tableau dashboard suite provides interactive features including:

- Year filtering
- Region filtering
- Country filtering
- Subject filtering
- University comparisons
- Interactive charts
- Geographic maps
- Dashboard navigation
- Cross-dashboard analysis

These features allow users to explore the dataset according to their specific analytical requirements.

14. Dashboard Navigation

The four dashboards are connected to create a unified analytical experience.

Navigation Flow

University Overview → Research Analytics → Student Analytics → Country Comparison

Users can move between different analytical perspectives without leaving the dashboard environment.

This allows the project to move from:

University-level analysis → Research analysis → Student analysis → Country-level benchmarking

15. Analytical Insights

EduVision_DV enables users to identify and analyze:

- Top-performing universities.
- University ranking differences.
- Academic reputation patterns.
- University distribution across countries and regions.
- Employment outcome differences.
- Research-performing institutions.
- Publication and citation patterns.
- Research productivity trends.
- International student representation.
- Student population differences.
- Student diversity trends.
- Faculty-to-student relationships.
- Country-level performance.
- Regional education trends.
- Top-performing countries.

16. Target Users

Students

The dashboard can help students:

- Compare universities.
- Explore university performance.
- Understand internationalization.
- Examine student diversity.

Researchers

Researchers can use the dashboard to:

- Compare research performance.
- Analyze publications.
- Examine citations.
- Study research productivity.

University Administrators

Administrators can use the dashboard for:

- Institutional benchmarking.
- Performance comparison.
- Student analysis.
- Research analysis.

Policymakers

Policymakers can use country and regional comparisons to understand higher-education trends.

Education Consultants

Consultants can use the dashboard for university and country benchmarking.

17. Data Quality and Validation

Data quality is an important part of the project.

The following areas are validated:

- Duplicate records
- Missing values
- University-name consistency
- Country-name consistency
- Ranking calculations

- KPI calculations
- Performance Index
- Dashboard filters
- Dashboard navigation
- Geographic visualizations
- Time-series trends
- University comparisons

The project also targets high dataset completeness and reliable dashboard calculations.

18. Project Evaluation

The project is evaluated across several areas.

Area	Expected Outcome
Data Collection & Cleaning	High-quality and sufficiently complete dataset
KPI Development	Multiple meaningful higher-education KPIs
Dashboard Development	Four integrated dashboards
Dashboard Functionality	Interactive charts, filters and navigation
Documentation	Complete and portfolio-ready documentation

19. Technology Stack

Technology	Purpose
Python	Data preparation and transformation
Pandas	Data analysis and manipulation
Scikit-learn	Feature normalization
Plotly	Exploratory visualization

Tableau Desktop	Dashboard development
Tableau Public	Dashboard publishing
Google Colab	Development environment
CSV / XLSX	Data storage
GitHub	Project repository and documentation

20. Project Deliverables

The final project includes:

1. Cleaned and integrated university dataset.
2. Final Tableau-ready dataset.
3. Four Tableau dashboards.
4. Tableau workbook.
5. Project README.
6. Project documentation.
7. GitHub repository.
8. Optional Tableau Public deployment.

21. Recommended GitHub Structure

EduVision_DV/

|

|— data/

| |— qs-world-rankings-2025.csv

| |— THE World University Rankings 2016-2026.csv

| |— university_cleaned.csv

| |— university final dataset.csv

```
|
|
|— dashboard/
|   └── EduVision_DV.twbx
|
|
|— docs/
|   └── project_documentation
|
|
└── README.md
```

22. Methodological Notes and Limitations

The following points should be considered when interpreting the dashboard:

- The **Performance Index is a project-specific measure** and is not an official QS or THE score.
- The **Global Rank Score** is a custom transformation used for analytical comparison.
- Range-based population and percentage values are converted into representative numerical values, which is an approximation.
- QS and THE datasets have different structures and coverage.
- Not every university record necessarily has a direct counterpart across both datasets.
- Dashboard conclusions depend on the available fields and data coverage.
- KPI comparisons should be interpreted within the methodology used in this project.

23. Future Enhancements

The project can be further improved by:

- Adding additional university-ranking datasets.
- Adding more ranking years.
- Improving university-name matching.
- Adding institution-level drill-down analysis.

- Adding year-over-year KPI changes.
- Adding dynamic KPI selection.
- Automating data refresh.
- Adding automated data-quality reports.
- Adding more research-impact indicators.
- Expanding country and regional benchmarking.
- Publishing the dashboard as a portfolio project.

24. Conclusion

EduVision_DV – Higher Education Performance Dashboard provides an integrated analytical view of global higher education.

The project combines university ranking information with research, student, academic, institutional, and geographical indicators.

The four dashboards provide complementary perspectives:

- **University Overview** – overall institutional performance
- **Research Analytics** – research output and impact
- **Student Analytics** – student and faculty-related analysis
- **Country Comparison** – international and regional benchmarking

The project demonstrates an end-to-end analytics workflow covering data preparation, data integration, KPI development, performance benchmarking, Tableau visualization, dashboard interactivity, testing, and documentation.

Overall, EduVision_DV provides a practical and interactive solution for exploring global university performance and higher-education trends.